Venkata Lakshmi Parimala Pasupuleti

Software Engineer · AI Integration & Production Evaluation · LLM-Powered Applications · Scalable Backend Systems · CI/CD · Test Automation · Observability · Python · AWS

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PROFESSIONAL SUMMARY

Software Engineer with 8 years shipping scalable production systems and two shipped AI integrations - AgentProbe and LeaseI - purpose-built to solve what this role targets: launching AI into production applications and evaluating their impact with appropriate metrics. AgentProbe is a 15-endpoint production AI evaluation platform with a 5-metric quality engine (correctness, groundedness, latency, token efficiency, hallucination rate), LLM-as-judge auto-rater, and CI-integrated regression gate - instrumented from day one to measure AI performance and block regressions before merge. LeaseI is an 8-agent LangGraph agentic system with HITL escalation and 93.6% precision across 1,240 test cases. At Entrata, owned scalable backend architecture at 99.5% uptime across 100K+ monthly active users, built CI/CD and test automation pipelines, and mentored 5+ engineers on engineering best practices. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Engineering & Backend: Python, SQL, FastAPI, PHP, JavaScript, TypeScript, Bash, Docker, Microservices, Scalable Distributed Systems, AWS (S3, EC2, RDS, Glue), Redis, RabbitMQ, REST APIs, CI/CD (GitHub Actions), Test Automation, Integration Testing, Reliability Engineering, PostgreSQL, MySQL, Oracle, Linux, Git

AI Integration & Evaluation: AI Integration, LLM Prompting, LLM-as-Judge, Production AI Evaluation, Model Performance Monitoring, Hallucination Detection, Observability, LangGraph, LangChain, Agentic AI, RAG Pipelines, ChromaDB, OpenAI API, Anthropic API, HITL, Prompt Engineering, MCP, FastAPI

ML & Data: Machine Learning, AutoML, PyTorch, scikit-learn, NLP, BERT, PySpark, Feature Engineering, Statistical Modeling, Tableau, Agile/Scrum

EXPERIENCE

Magizhs Tech Solution
May 2025 - Aug 2025

AI/ML Engineer Intern

- Launched AI integration into a production mission-planning system (LangGraph agentic backend): end-to-end from LLM prompting and NLP ingestion through conditional routing and validated output - evaluated impact with test automation across 100% of compliance scenarios before production deployment.
- Implemented RAG pipeline with production observability on retrieval quality; monitored AI performance in production and adjusted prompting strategy based on groundedness metrics.

George Mason University
Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built ML pipelines (Python, R) on WHO and UK Road Safety datasets - $R^2 = 0.977$; full pipeline from data ingestion through feature engineering, model optimization, and statistical validation across 190+ countries.
- Coached 600+ students in Python, SQL, and database design on AI/ML best practices; designed dashboards improving research comprehension by 40%.

Entrata - PropTech / Affordable Housing SaaS
Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Architected and delivered 20+ scalable production modules end-to-end** - drove full software lifecycle from design through CI/CD, test automation, and production monitoring across 120+ Agile sprints; 99.5% uptime SLA at 100K+ monthly active users; zero compliance violations across 7 years.
- **Built CI/CD and test automation infrastructure** - integration and functional testing pipelines eliminating manual verification across compliance-critical workflows; reliability engineering maintaining sub-10 bug count for 6+ consecutive months; observability tooling reducing triage time 40%.
- **Designed scalable distributed backend architecture** - Docker microservices redesign (50% scalability improvement, 30% deployment reduction); RabbitMQ event-driven processing with retry logic and idempotency; Redis caching for high-performance data access.
- **Cloud infrastructure and database optimization at scale** - AWS S3, EC2, RDS; 200+ SQL query optimizations (30% performance gain); scalable data processing pipelines (fragmented federal data into 180,000 domain records per release cycle); 100% PHI compliance.
- Mentored 5+ junior engineers on software engineering best practices and AI tooling adoption; 300+ peer code reviews; drove cross-team technical alignment.

Entrata India - Pune
Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built property management backend modules (PHP, Python, PostgreSQL); scalable data pipeline for 1.6M HUD/Tax Credit records; AWS S3 migration resolving 90% of legacy storage constraints.

PROJECTS

AgentProbe: Production AI Integration & Evaluation Platform AI Integration · Eval

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) | github.com/venkatalakshmiparimala/AgentProbe

Launched AI integration into production and evaluated its impact with appropriate metrics: 15-endpoint FastAPI backend, 5-metric quality engine (correctness, groundedness, latency, token efficiency, hallucination rate), LLM-as-judge auto-rater measuring AI performance in production, CI-integrated regression gate blocking quality regressions before merge, and multi-provider LLM routing. Used to evaluate LeaseI across 1,240+ integration test cases - zero regressions shipped.

LeaseI: 8-Agent AI Integration - Agentic Investigation Platform AI Integration · Agentic

Python · LangGraph (8-agent) · FastAPI · TF-IDF · OCR · SQLite | github.com/venkatalakshmiparimala/Lease_complaince_rag

AI integration into a production compliance investigation platform: 8-agent LangGraph stateful system with LLM prompting, conditional routing, and HITL escalation when model confidence is below threshold. 93.6% precision, 87.9% F1 across 1,240 test cases; CI quality gate with telemetry-driven AI performance scoring.

AI on FHIR: NLP-Driven Medical Records Query System NLP · LLM

Python · Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings | github.com/venkatalakshmiparimala/AI_on_FHIR

AI integration using LLM prompting and BERT embeddings to translate natural-language queries into FHIR API calls - reduced retrieval time ~60%; HIPAA-compliant RBAC across structured medical records with full data isolation.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017